

CSC-350 Artificial Intelligence Course Project

Group Contribution Statement

We, the undersigned, hereby declare the individual tasks performed and the contributions made by each group member towards the successful completion of our AI Semester Project entitled 'Advanced Time-Series Forecasting using Lorentzian Distance Classification & Kernel Regression'.

Tayyab Mangi (CMS: 023-24-0118)

- Coded the core machine learning training, cross-validation, and backtesting scripts.
- Implemented the Lorentzian distance custom metric within Scikit-Learn's KNeighborsClassifier.
- Wrote the methodology, results, and analysis sections of the project report and created the presentation slides.

Asif Ali Rattar (CMS: 023-24-0158)

- Wrote the Python code to fetch historical data from Binance Klines.
- Implemented the technical indicators (RSI, WaveTrend, CCI, ADX) and the Nadaraya-Watson regression calculations.
- Generated the evaluation plots (confusion matrix, ROC curve, and backtest results chart) and drafted the introduction and literature review sections of the report.

Tayyab Mangi

Date: 23rd May 2026

Asif Ali Rattar

Date: 23rd May 2026